

Contract XLSB health and performance audit

Authoritative master: TestFileClean.xlsxb

Functional integrity, calculation architecture and optimisation roadmap

Audit date: 24 August 2026

Status: Evidence complete - no workbook changes made

Overall assessment

The workbook is structurally healthy and natively fast. Prioritise cross-engine correctness and controlled robustness improvements; do not undertake a wholesale workbook rewrite.

Executive summary

The authoritative master and repository compatibility copy are byte-for-byte identical. The saved workbook passes every populated Check Sheet status, has no broken defined names, external links, data connections, circular reference or full-column calculation formulas, and recalculates quickly in native Excel on the audit machine.

The main confirmed defect is in Summit's PMCOST custom-function metadata, not in the XLSB formula call. The main workbook risk is the large number of 3-D references whose meaning depends on worksheet ordering. Current speed does not justify broad formula replacement.

638,360	2.15 s	35,920	0
Formula cells	Full dependency rebuild	3-D reference formulas	Broken names / external links

Decision: keep the workbook-first architecture, preserve formula-backed Transactional DB and embedded VBA, and focus engineering effort on UDF parity, sheet-order guards, calculation instrumentation and measured micro-optimisations.

Priority findings

ID	Priority	Finding	Recommended action
F1	High	Summit PMCOST metadata does not match the eight-position compatibility signature	Preserve the eight-position compatibility signature, document that FinalYear is unused, advertise six required value positions plus two required references, and add parity tests before relying on Summit recalculation.
F2	High	Cross-engine UDF parity is not regression-tested	Create an automated parity harness comparing the 760 call results and key outputs after Excel/VBA and DevExpress recalculation. Make RESPCOST parameters required if they are unconditionally dereferenced and add explicit error guards.
F3	Medium	35,920 formulas depend on worksheet ordering through 3-D references	Retain these efficient formulas for now, but add an expected sheet-order/block-membership manifest and validate it on open, save, and structural commands.
F4	Medium	Custom functions perform repeated formula-engine MATCH evaluations	Replace repeated Engine.Evaluate calls with direct range reads and an indexed year-to-rate lookup, preserving Excel MATCH semantics and testing parity.
F5	Medium	Formula complexity is concentrated and highly repetitive	Refactor only measured hotspots using helper rows/columns or cached match indices. Do not introduce LET/XLOOKUP until the minimum supported Excel version is agreed.
F6	Low-Medium	Volatile CELL calls are duplicated	Evaluate each CELL result once in a helper cell (or LET only if supported) and reuse it, reducing the menu occurrences by about 75%.
F7	Low-Medium	Several sheets have inflated used ranges	On a disposable copy, remove only demonstrably obsolete formatting/objects beyond the real model area, then run full Excel/VBA/Summit round-trip tests.
F8	Low	One unguarded #N/A remains outside intentional chart suppression	Confirm the expected blank-state behaviour and use IFNA/validation if absence from ImportedSchemes is normal.

Scope, authority and evidence controls

Control	Result
Authoritative master	Z:\Sandbox\TestFileClean.xlsb
Repository copy	Z:\Repos\Abovo Summit\Library\TestFileMigrated.xlsb
Exact-copy verification	PASS - 11,841,039 bytes; SHA-256 B248B1C733E1E3293536FBE1DBC9576D56FD4D34BEB74A3898B7F3D7333BCFBE
Excel inspection	Disposable exact local copy; read-only; macros/events disabled; links not updated; never saved
Protected sheets	35 unlocked only in memory on the disposable copy; password not persisted
VBA handling	Project presence confirmed; source neither extracted nor executed
Timing boundary	Native Excel only; Summit calculation/UI refresh excluded

This audit deliberately preserves the standalone Excel/VBA contract. Recommendations that alter formulas, names, used ranges, worksheet order, protection or visibility require a disposable-copy regression before adoption.

Structural health

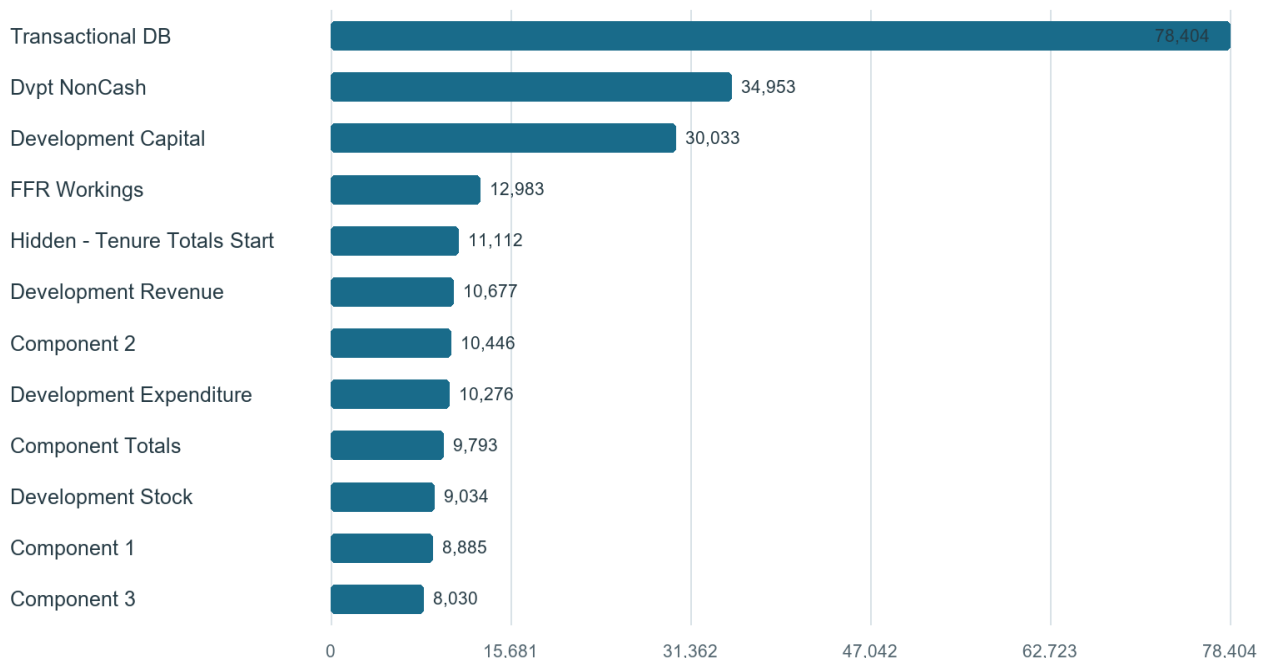
Area	Evidence	Assessment
Workbook	281 sheets; XLSB format 50; embedded VBA present	Contract intact
Defined names	1,758 total; 262 multi-area; max 28 areas	No broken/external/volatile names
Dependencies	638,360 formulas; 5,625 R1C1 patterns	Highly repetitive, maintainable with controls
External state	0 link sources; 0 connections; 0 circular references	Strong standalone baseline
Validation	All saved populated Check Sheet statuses are OK	Pass at saved baseline
Presentation	36 charts; 1,076 shapes; 947 conditional-format rules	Feature-rich; potential render overhead

Calculation performance

Test	Result	Interpretation
Read-only open	2.37 s	Healthy for an 11.8 MB / 281-sheet model
Warm CalculateFull #1	0.43 s	Fast
CalculateFullRebuild	2.15 s	Fast dependency rebuild
Warm CalculateFull #2	0.41 s	Stable
Calculation mode	Automatic; calculate-before-save enabled	Appropriate; no forced full-load calculation

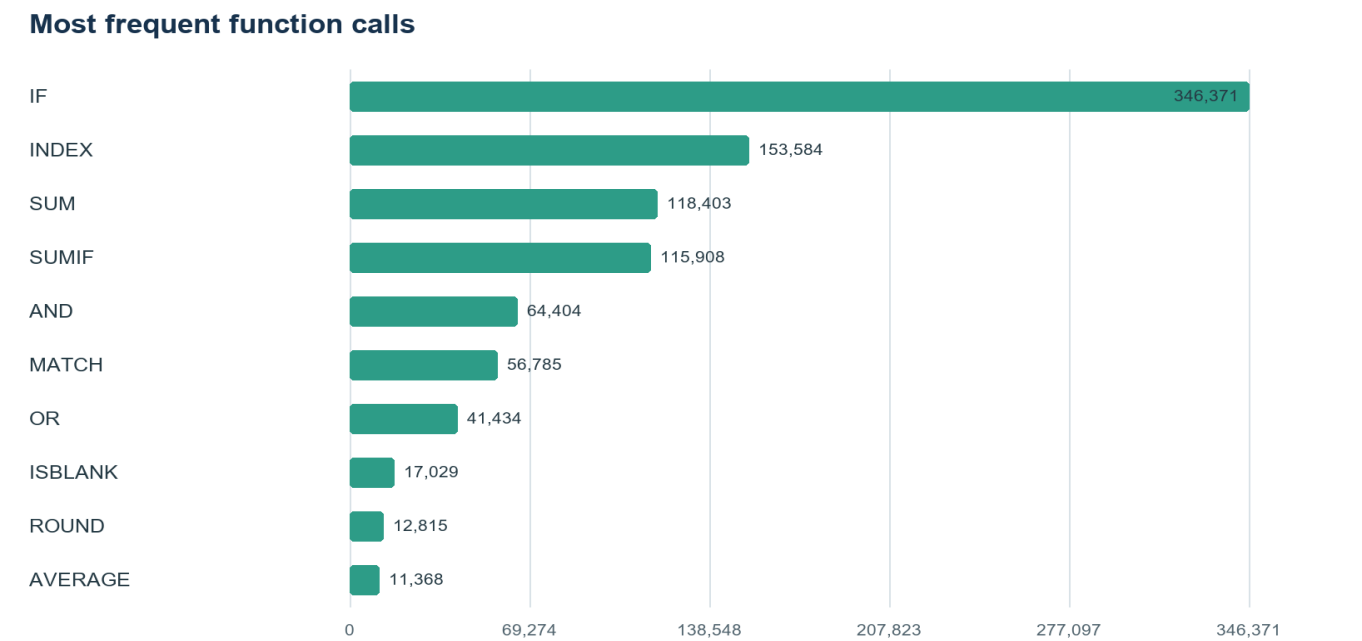
The current workbook does not have a general native-Excel speed problem on the audit machine. A broad formula rewrite would carry more correctness risk than demonstrated benefit. Summit-side timings must be instrumented separately because DevExpress custom functions, RangeDataSource refresh and UI update are outside these measurements.

Formula concentration by worksheet



Formula concentration: Transactional DB alone is 12.3% of formulas; the top 10 sheets are 34.1% and the top 40 are 62.9%.

Formula architecture



Metric	Count	Observation
Formula cells	638,360	Large but efficiently cached/recalculated
Distinct R1C1 patterns	5,625	High reuse across time series
500+ character formulas	6,582	Maintenance hotspot
Maximum formula length	901	No formulas exceed 1,000 characters
3-D reference formulas	35,920	Efficient but sheet-order dependent
Full-column formula refs	0	Good bounded-range discipline
Volatile CELL occurrences	1,313	Concentrated in menu/list sheets

3-D reference dependency risk

The model uses 35,920 3-D formulas to aggregate sheets between named worksheet boundaries. This is performant and concise, but the dependency is positional: moving a sheet can change totals without creating #REF!.

Observed pattern	Risk	Control
SUM('Component 1:Component 12'!...)	Component sheet inserted outside boundary	Validate exact component membership/order
SUM('Journals:Total BSheet Adj TB'!...)	New adjustment sheet omitted or unrelated sheet included	Validate boundary block on open/save
SUM('Hidden - Tenure Totals Start:...End'!...)	Development insert/move changes grouped totals	Structural commands must assert block integrity

Custom-function contract and Summit parity

Confirmed issue: the workbook's representative PMCOST formula passes eight compatibility arguments, but FinalYear is deliberately unused, so seven inputs affect the result. Summit's `PMCostFunction` advertises seven `ParameterInfo` entries despite accepting an eight-position call.

Function	Workbook calls	Workbook args	Summit metadata	Summit evaluation
PMCOST	380	8 positions; 7 used	7 entries	Reads 0-7; FinalYear unused
RESPCOST	380	7	1 required + 6 optional	Unconditionally reads all 7

Both Summit functions execute `FormulaEngine.Evaluate(MATCH(...))` within a loop. Replacing this with a direct indexed lookup is likely the best calculation-speed opportunity in Summit, but only after exact parity tests for approximate MATCH semantics, missing years, invalid ranges and division-by-zero cases.

Required regression: calculate the same disposable workbook in macro-enabled Excel and Summit, then compare all 760 UDF result cells, Check Sheet, Development Expenditure totals, statements and Transactional DB. This audit did not execute VBA and therefore cannot certify UDF equivalence.

Saved errors and validation

Observation	Count	Assessment
Saved formula errors	467	466 deliberate chart #N/A gaps; 1 unguarded lookup
Saved populated Check Sheet statuses	All OK	Baseline passes
Macro-disabled rebuild Check Sheet errors	7 #NAME?	Audit limitation; not a workbook defect
External links / connections / circular refs	0 / 0 / 0	Strong

Confirm whether Hidden - Tenure Totals Start!B6 should be blank when no imported scheme matches. If yes, guard the INDEX/MATCH with IFNA or an explicit import-state check so the latent error cannot leak into future interfaces.

Used-range and presentation overhead

Worksheet	Used range dimensions	Last populated dimensions	Excess rectangle cells
5 Yr Monthly Cashflow	337 x 44	102 x 44	10,340
5 Yr Quarterly Cashflow	262 x 41	27 x 41	9,635
Transactional DB	1910 x 78	1909 x 74	7,714
Dvpt Component Depn	469 x 37	468 x 23	6,589
Component 1	1695 x 15	1693 x 13	3,416
Component 2	1695 x 15	1693 x 13	3,416
Component 3	1695 x 15	1693 x 13	3,416
Component 4	1695 x 15	1693 x 13	3,416
Component 5	1695 x 15	1693 x 13	3,416
Component 6	1695 x 15	1693 x 13	3,416
Component 7	1695 x 15	1693 x 13	3,416
Component 8	1695 x 15	1693 x 13	3,416

These counts identify candidates, not safe deletions. Some excess areas hold print formatting, future structural templates, validations, chart anchors or Summit expansion geometry. Cleanup should be attempted only on a disposable copy and accepted only after Excel/VBA/Summit round-trip and structural Add Lines tests.

Feature	Count	Concentration / note
Visible worksheets	279	Navigation and accidental-edit exposure; hide only after client workflow review
Shapes	1,076	148 on Transactional DB; 80 on FFR Workings; many are likely navigation/controls
Charts	36	Concentrated on seven dashboard/output sheets
Conditional-format rules	947	79 on Development BP Assumptions
Validation cells	16,053	4,163 on Development BP Assumptions

Recommended implementation sequence

Phase	Action	Expected value	Gate
0 - Baseline	Automate Excel/VBA vs Summit output comparison and calculation timing	Prevents silent parity regressions	No workbook change
1 - Correctness	Preserve PMCOST's 8-position signature, correct its metadata, make RESPCOST explicit and add guards	Removes confirmed cross-engine risk	760-cell UDF parity
1 - Performance	Replace nested Engine.Evaluate MATCH loops with direct indexed lookup	Largest likely Summit speed gain	Exact value/error parity
2 - Structure	Validate every 3-D sheet block and multi-area name after structural commands	Prevents silent total changes	All 12 structural domains
2 - Low risk	Consolidate volatile CELL calls	Small recalc improvement	Excel version compatibility
3 - Hygiene	Trial used-range/obsolete-shape cleanup	Smaller render/navigation footprint	Full visual/round-trip test
3 - Refactor	Refactor only profiled long/repeated formulas	Auditability and selective speed	No output drift

Minimum regression pack

1. Exact master/library hash and XLSB/VBA preservation.
2. Macro-enabled Excel full calculation followed by every populated Check Sheet status.
3. Summit recursive calculation of the same copy and cell-by-cell comparison of 760 UDF results.
4. Transactional DB A6:BV1599 and all named mirrors reconciled after add/delete operations.
5. Expected worksheet membership/order for every 3-D boundary block.
6. All 12 structural rule domains, especially Development and Capital Grant multi-range changes.
7. Summit save -> Excel/VBA open/save/recalculate -> Summit reopen.
8. Five-run timing split into calculation, UDF, datasource refresh and UI refresh on client hardware.

Appendix A - top formula worksheets

Rank	Worksheet	Formula cells	% of workbook	Long formulas
1	Transactional DB	78,404	12.28%	0
2	Dvpt NonCash	34,953	5.48%	0
3	Development Capital	30,033	4.70%	2,850
4	FFR Workings	12,983	2.03%	0
5	Hidden - Tenure Totals Start	11,112	1.74%	0
6	Development Revenue	10,677	1.67%	0
7	Component 2	10,446	1.64%	0
8	Development Expenditure	10,276	1.61%	2,090
9	Component Totals	9,793	1.53%	0
10	Development Stock	9,034	1.42%	0
11	Component 1	8,885	1.39%	0
12	Component 3	8,030	1.26%	0
13	Component 4	8,030	1.26%	0
14	Component 5	8,030	1.26%	0
15	Component 6	8,030	1.26%	0
16	Component 7	8,030	1.26%	0
17	Component 8	8,030	1.26%	0
18	Component 9	8,030	1.26%	0
19	Component 10	8,030	1.26%	0
20	Component 11	8,030	1.26%	0
21	Component 12	8,030	1.26%	0
22	Hidden - Funds Available	7,339	1.15%	193
23	Dvpt Component Depn	7,033	1.10%	0
24	Dvpt Imp FFR Info	6,622	1.04%	0
25	All Schemes FFR	5,962	0.93%	0
26	Development Inflation Factors	5,341	0.84%	0
27	Dvpt Factors	5,287	0.83%	0
28	Total Non Cash Trial Balance	5,276	0.83%	0
29	Hidden - Investments	5,215	0.82%	0
30	Hidden - Year Table	5,067	0.79%	0

Appendix B - defined-name topology

Metric	Count	Comment
Total names	1,758	All range-backed
Workbook scoped	1,676	Primary model API
Worksheet scoped	82	Mostly print/filter names
Hidden	15	No broken references
Multi-area	262	Structural expansion sensitivity
Maximum areas	28	FundDvptDates
Full-column / full-row names	36 / 20	Largest are worksheet print titles, not calculation formulas

Appendix C - evidence limits and interpretation

- Timings are point measurements on the audit workstation; repeat on representative client hardware before setting service levels.
- The workbook was inspected through Excel COM with macros/events disabled. VBA project presence was confirmed, but no VBA source was extracted or executed.
- A macro-disabled full rebuild cannot evaluate workbook VBA UDFs. Seven resulting #NAME? Check Sheet rows are recorded as an evidence limitation, not a defect.
- Per-sheet Calculate calls were dominated by COM overhead and are excluded from performance ranking.
- No code or workbook remediation was implemented. Every recommendation requires a separately approved change and regression pack.

Persistent evidence files: Library/Contract_XLSB_Audit_2026-08-24.md and Library/Contract_XLSB_Audit_Evidence_2026-08-24.json. Future workbook/calculation tasks must review them together with the existing project scope audit and index.